When Detectors Don't Steer: A Four-Gate Audit of Activation Interventions in Gemma-3-4B-IT

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Abstract

Predictive internal signals are often used as steering targets, although readout quality alone does not establish behavioral control. A single-model Gemma-3-4B-IT case study audits this inference across contextual faithfulness, multiple-choice answer selection, open-ended QA, and jailbreak evaluation. On FaithEval, H-neurons and sparse autoencoder (SAE) features achieve similar held-out readout quality (AUROC 0.843 vs. 0.848), yet only H-neuron scaling produces a reliable behavioral dose-response under the tested operators (+2.1 pp/α [1.4, 2.8]). Within the same Gemma SAE candidate pool, prompt-end utility and answer-span selectors produce metric-specific margin shifts without recovering the compliance endpoint. ITI improves TruthfulQA multiple-choice answer selection while reducing open-ended TriviaQA bridge accuracy, most often through wrong-entity substitution. Jailbreak conclusions depend on scoring granularity and evaluator construct. These results support a gate-based audit that separates measurement, localization, control, and externality.

1. Introduction

A predictive internal signal (a neuron, feature, or direction that reliably discriminates between behavioral categories on held-out data) is a tempting steering target. If a feature predicts whether a model will hallucinate or comply harmfully, it is natural to expect that amplifying or suppressing it will steer behavior. This heuristic underlies much recent work, from representation engineering and activation addition to inference-time intervention and neuron-level steering (Zou et al., 2023; Turner et al., 2023; Li et al., 2023; Gao et al., 2025; Arditi et al., 2024).

The heuristic sometimes works. Refusal-mediating directions produce reliable refusal modulation (Arditi et al., 2024), and hallucination-associated neurons can shift over-compliance (Gao et al., 2025). The heuristic also sometimes fails, and the failure modes have received less systematic attention than the successes.

In Gemma-3-4B-IT (Google DeepMind, 2025), the readout-to-steering heuristic shows repeated dissociations between four stages that are routinely conflated:

1. **Measurement.** Is the evaluation trustworthy? Scoring granularity changed the scientific conclusion, while output-length checks and the evaluator audit exposed residual measurement risk and rubric dependence (§5).
2. **Localization.** Does the readout identify causally relevant components? SAE features matched H-neurons on detection quality but did not translate into useful control (§3).
3. **Control.** Does intervention produce the intended behavioral change? Effects were narrow and task-local (§4).
4. **Externality.** Does the effect transfer without causing harm? Wrong-entity substitution was the most common manually coded failure mode on a held-out bridge benchmark (§4).

Contributions. (1) On Gemma FaithEval, H-neurons and SAE features reach similar held-out readout quality, but only H-neuron scaling produces reliable control under matched surface evaluation. (2) Within the tested Gemma SAE candidate pool, readout, prompt-end utility, and answer-span utility selectors produce metric-specific margin effects, and all miss the compliance endpoint. (3) TruthfulQA-fitted ITI improves MC answer selection but reduces TriviaQA bridge accuracy, with wrong-entity substitution dominating the manually coded failures. (4) In the jailbreak evaluator setting, scoring granularity changes verdicts, while output-length exposure and evaluator construct choices remain reporting risks. (5) The four-gate scaffold separates measurement, localization, control, and externality claims.

Study orientation. The study is a single-model comparative case study in Gemma-3-4B-IT (Google DeepMind,

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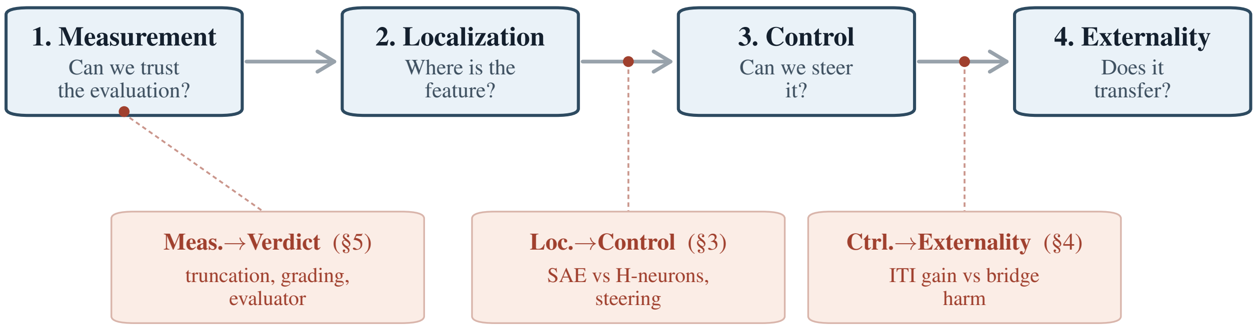

Figure 1. Four-stage scaffold for activation-intervention claims. Measurement, localization, control, and externality are distinct stages, and each transition requires separate evidence. In this case study, measurement choices changed the jailbreak verdict (§5), matched readouts diverged under control (§3), and successful TruthfulQA MC control failed on open-ended generation (§4).

2025). The comparison covers multiple intervention families (neuron scaling, SAE feature steering, and inference-time intervention via attention heads) across contextual faithfulness (FaithEval (Ming et al., 2025)), answer selection (TruthfulQA (Lin et al., 2022)), open-ended factual generation (TriviaQA (Joshi et al., 2017), SimpleQA Verified (Haas et al., 2025)), domain QA (BioASQ (Tsatsaronis et al., 2015)), and jailbreak settings (JailbreakBench (Chao et al., 2024)). All claims are surface-specific: FaithEval is a context-faithfulness / anti-compliance surface, TruthfulQA MC constrained answer selection, TriviaQA bridge and SimpleQA Verified open-ended factual generation, and JailbreakBench a harmfulness-evaluation surface. Within that scope, the empirical questions are: (1) Is held-out readout quality enough to select steering targets? (§3) (2) Do answer-selection gains transfer to open-ended factual generation? (§4) (3) Do plausible measurement choices leave the jailbreak verdict stable? (§5) They do not claim a new steering method or a general failure of detector-based targets. Supplement stress tests on `Mistral-Small-24B-Instruct-2501` and SIMID diagnostic artifacts are used only as failed or incomplete gates.

2. Related Work

Foundational probe critiques establish that high readout accuracy can reflect nonfunctional correlates (Hewitt & Liang, 2019; Elazar et al., 2021; Kumar et al., 2022). The linear representation hypothesis (Park et al., 2024) makes the complementary positive prediction that concepts encoded as linear directions should support both detection and control, which the Gemma experiments probe empirically. Hase et al. (2023) provide a close editing precedent: better localization can fail to predict better control.

The steering lineage runs from representation engineering and activation addition (Zou et al., 2023; Turner et al., 2023) through ITI (Li et al., 2023). Within SAE-based interventions, foundational feature work (Cunningham et al., 2023) and the Gemma Scope release (Lieberum et al., 2024) make sparse features available at scale, while target-selection studies show limits of salience as a steering criterion: high-scoring SAE features can fail to control behavior (Arad et al., 2025), detection and steering can require different objectives (Wu et al., 2025), Wang et al. (2026) report only a weak positive association between SAE interpretability and steering utility across 90 SAEs (Kendall's $\tau_b \approx 0.298$), and Bhalla et al. (2024) describe a predict/control discrepancy for interpretability-based interventions.

Transfer and reliability work narrows the claim space further. Li et al. (2023) already showed that ITI's multiple-choice gains can diverge from generative gains; Tan et al. (2024) study the brittleness and input sensitivity of steering vectors; Pres et al. (2024) show that behavior-steering evaluations can be format-dependent; and Opiełka et al. (2026) show that causally effective vectors can still be surface-specific. Positive counterexamples remain important: refusal directions and targeted multi-attribute steering can work when the behavioral target is better matched to the intervention (Arditi et al., 2024; Nguyen et al., 2025).

Measurement fragility is a parallel reviewer concern. JailbreakBench (Chao et al., 2024) provides the benchmark surface used here; StrongREJECT (Souly et al., 2024) showed that binary scoring can overestimate jailbreak effectiveness; Chen & Goldfarb-Tarrant (2025) and Eiras et al. (2025) show that evaluator artifacts and rubric choices matter; and Huang et al. (2025) argue that guideline-conditioned judging can materially change jailbreak conclusions. For factuality more directly, Yona et al. (2026) argue that hallucination mitigation should report utility-error tradeoffs and spillover costs alongside calibration or single operating points; the

SimpleQA and bridge analyses instantiate those cautions for activation interventions.

Concurrent SAE-centric analyses (Arad et al., 2025; Wang et al., 2026) and synthetic-benchmark evaluations (Wu et al., 2025) make detection/control divergence increasingly clear within single-family settings. The cross-representational FaithEval comparison evaluates neurons and SAE features on the same real behavioral surface with closely matched held-out readout quality. Its scope is Gemma-local and limited to the tested operators; basis, layer-coverage, and operator-form differences remain open, so it is not an operator-matched proof that neurons are better causal bases than SAE features. Two supporting additions broaden that case study. First, TriviaQA bridge generation provides a frozen-protocol behavioral diagnosis of the transfer failure, where wrong-entity substitution is the most common manually coded right-to-wrong pattern and formal refusal is rare. Second, scoring granularity can change the verdict for the same representation-engineering intervention outputs, while output-length checks and a held-out evaluator audit reveal residual measurement risk. Together these observations support a four-stage audit framework (measurement, localization, control, externality).

3. Similar Readout Quality Does Not Guarantee Control

3.1. The Readouts Are Real

The intervention targets were selected through held-out predictive readouts. A sparse H-neuron probe following Gao et al. (2025) identified 38 positive-weight neurons out of 348,160 feed-forward neurons and achieved AUROC 0.843 (accuracy 76.5%, $n_{\text{test}} = 780$). A parallel L1 probe on Gemma Scope 2 SAE activations (Lieberum et al., 2024) selected 266 positive-weight features across 10 layers and achieved AUROC 0.848 (accuracy 77.2%, $n_{\text{test}} = 782$). These are real held-out signals, though individual-weight interpretation remains fragile.

3.2. Comparable Readouts, Divergent Control

Both methods achieve similar held-out detection quality, and both interventions are evaluated on the same 1,000 FaithEval items (Ming et al., 2025). The 38 probe-selected neurons were scaled multiplicatively across $\alpha \in \{0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, with $\alpha = 1.0$ as the no-op baseline. The 266 tested Gemma Scope SAE features were scaled through an encode-modify-decode intervention on the same grid.

The neuron intervention showed a clear dose-response: compliance slope $+2.09$ pp/α $[1.38, 2.83]$, with perfect Spearman monotonicity ($\rho = 1.0$). Relative to the no-op baseline, compliance at $\alpha = 3.0$ increased by $+4.5$ pp $[2.9, 6.1]$. Across five unconstrained and three layer-matched random-neuron sets, all eight random seeds produced null slopes (maximum $+0.21$ pp/α); every paired neuron-minus-random difference excluded zero.

The tested SAE intervention did not produce the same behavior. The full-replacement H-feature slope was $+0.16$ pp/α $[-0.51, 0.84]$, with no monotonic trend ($\rho = 0.18$). A delta-only architecture that cancels reconstruction error gave near-zero slopes: H-feature $+0.12$ pp/α and random-feature -0.09 pp/α (audit summary; no slope CI in the package). The full-replacement SAE curve in Figure 2 includes non-monotone path/reconstruction movement shared by target and random SAE sets; the claim-bearing feature-specific check is the delta-only contrast. The neuron baseline on the same three-alpha subset remained $+2.12$ pp/α.

The paired neuron-minus-SAE slope difference was $+1.93$ pp/α $[+0.94, +2.92]$ (permutation $p < 0.001$). On the same 1,000 FaithEval items, similar held-out readout quality failed to predict intervention utility: H-neurons and SAE features showed comparable detector performance; only the neuron intervention produced a robust behavioral dose-response.

Within-SAE target-selection ablation. A follow-up ablation tested whether the SAE null could be explained by a poor target-selection rule. The ablation used the same 509-feature Gemma SAE candidate set, the same extraction layers, and a single $\alpha = 0$ delta-only operator (feature zeroing) on a frozen $160/840$ validation/test FaithEval split. The first ablation compared top-266 readout features, top-266 prompt-end validation-utility features, and ten layer-matched activation-weighted random zero-weight controls; all produced null compliance deltas on the $n = 840$ test split (all paired 95% CIs include zero; point estimates ≤ 0.72 pp). On the anti-compliance log-prob margin, readout-selected features shifted the margin in the *wrong* direction (+0.92 nats $[+0.61, +1.23]$ vs. no-op), while prompt-end utility-selected features reduced it by -0.76 nats $[-1.08, -0.42]$; the paired utility-minus-readout delta was -1.68 nats $[-2.17, -1.19]$. Because both sides of that contrast share the same $\alpha = 0$ delta-only operator, any path artifact cancels; a dedicated path-drift control ($\alpha = 0$ on a single provably-dead zero-weight feature) independently moved the margin by only $+8.2 \times 10^{-8}$ nats $[-2.4 \times 10^{-3}, +2.1 \times 10^{-3}]$. A later answer-span selector on the same pool added a narrower finding: it reduced its native first-3-token answer-text margin (-0.33 nats $[-0.61, -0.06]$ vs. no-op), but left compliance null and underperformed the prompt-end utility selector on the original prompt-end margin ($+0.46$ nats $[+0.16, +0.75]$). Within the same candidate pool, readout-, prompt-end-utility-, and answer-span-utility criteria therefore induce metric-specific margin effects, and no rule reaches the accuracy endpoint

(Appendix C).

3.3. Synthesis

H-neurons *did* steer FaithEval compliance, and specificity was confirmed against eight matched random controls. The refusal-direction literature (Arditi et al., 2024) and targeted multi-attribute steering (Nguyen et al., 2025) are positive cases. The narrower claim is that readout quality alone is an unreliable heuristic for predicting *when* detection-based targets will steer behavior. The tested SAE features matched H-neurons on the quantity most commonly used to justify steering (held-out AUROC) and missed the downstream behavioral endpoint.

This comparable-readout design still confounds representational basis with readout quality: SAE features and neurons differ in operator form, layer coverage, and feature granularity. The within-SAE ablation reduces the target-selection concern by holding operator form and available extraction layers fixed; readout-, prompt-end-utility-, and answer-span-utility selectors produce metric-specific margin effects, and none reaches the accuracy endpoint. The cross-family basis confound remains, so the result does not establish that neurons are better causal bases than SAE features. It shows that readout quality alone did not predict steering utility in either the cross-representational comparison or within the tested SAE pool. This limitation returns in §6.

4. Control Is Surface-Local and Can Externalize

4.1. Positive Results Are Task-Local

In these experiments, detector-selected interventions can steer behavior on their primary surfaces. H-neuron scaling (38 magnitude-ranked neurons) produces clear, dose-dependent effects on compliance-adjacent surfaces. On FaithEval, compliance increases by $+4.5$ pp $[2.9, 6.1]$ above the no-op baseline at $\alpha = 3.0$ (slope $+2.09$ pp/α), confirmed by 8-seed negative controls showing flat random-neuron baselines (mean slope $+0.02$ pp/α). On FalseQA (Hu et al., 2023) ($n = 687$), the same neurons show a dose-response slope of $+1.62$ pp/α $[0.52, 2.74]$.

However, the same 38 neurons show no robust net accuracy effect on BioASQ factoid QA (Tsatsaronis et al., 2015): -0.06 pp $[-1.5, 1.4]$ on a well-powered sample ($n = 1{,}600$, MDE $\sim$2 pp), despite substantially perturbing answer style in 1,339 of 1,600 responses. The flat accuracy endpoint coexists with substantial output perturbation, suggesting that behavioral activity and alias-accuracy movement come apart on this surface. The intervention modulates compliance-adjacent behavior but does not improve domain-specific factual accuracy under the alias metric.

4.2. ITI: Answer Selection vs. Open-Ended Generation

ITI with directions fit from TruthfulQA multiple-choice contrasts (Lin et al., 2022) following Li et al. (2023) improves answer selection. At $\alpha = 8.0$, it yields $+6.3$ pp MC1 $[3.7, 8.9]$ and $+7.49$ pp MC2 $[5.28, 9.82]$ on held-out TruthfulQA folds, with 61 incorrect$\rightarrow$correct flips against 20 correct$\rightarrow$incorrect flips.

This MC/generation divergence echoes the tradeoff concern already visible in Li et al. (2023); here, SimpleQA Verified serves as a supporting stress test, while the locked bridge benchmark below is the main externality result. The gain does not transfer to open-ended factual generation. On a local copy of the 1,000-question SimpleQA Verified benchmark (Haas et al., 2025), graded with the benchmark's verified autorater, and with a prompt that removes the explicit escape hatch, ITI at $\alpha = 8.0$ reduces correct-answer rate from 4.6% $[3.5, 6.1]$ to 2.8% $[1.9, 4.0]$ (paired $\Delta = -1.8$ pp $[-3.1, -0.6]$). The attempt rate collapses from 99.7% to 67.0%, while precision among attempted answers remains stable ($\sim$4%), suggesting epistemic caution without factual improvement. Answer-selection success therefore provides no evidence for open-generation factual improvement, consistent with Pres et al. (2024) and Opiełka et al. (2026).

4.3. Bridge: Wrong-Entity Substitution

The TriviaQA bridge benchmark (Joshi et al., 2017) (held-out test set, $n = 500$, baseline adjudicated accuracy 45.0%) provides the main externality result: a specific behavioral failure mode beyond the score drop. The test set was evaluated under a one-shot frozen protocol: all parameters were locked from Phase 2 dev results, and no parameters were tuned on test data.

The intervention changes outputs beyond suppression. At $\alpha = 8.0$, the baseline TruthfulQA-sourced ITI variant reduces adjudicated accuracy by -5.8 pp $[-8.8, -3.0]$ (CI excludes zero), with 43 right-to-wrong flips against 14 wrong-to-right rescues (net -29, McNemar $p = 0.0002$). Both the primary (adjudicated) and floor (deterministic) accuracy deltas exclude zero, ruling out a grading artifact.

Refusal is a minor part of the observed harm. NOT_ATTEMPTED counts rise from 2 to 8, a small 1.2 pp effect. Formal refusal accounts for 0 of 43 R$\rightarrow$W flips under the author-plus-LLM adjudication. The dominant damage is factual corruption.

Wrong-entity substitution is the most common adjudicated failure mode. Entity-level factuality work treats wrong-entity substitution as a recognizable neighboring error family in generation (Nan et al., 2021). That literature supplies adjacent taxonomy only; it provides no prior evidence for this exact steering failure mode. One author and a blinded GPT-4o judge independently coded each of the

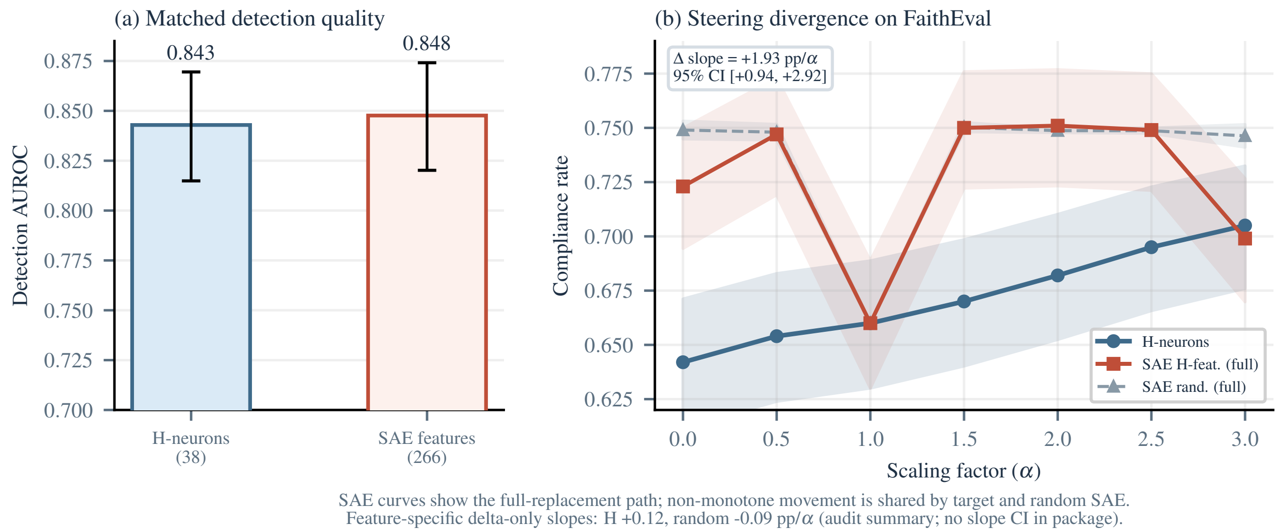

Figure 2. The anchor result: FaithEval comparable readouts, divergent control. (a) H-neurons and SAE features have similar held-out detection AUROC. (b) H-neurons produce a dose-dependent compliance increase. The SAE curves are full-replacement paths with non-monotone movement shared by target and random SAE sets; feature-specific delta-only slopes remain near zero (H $+0.12$, random -0.09 pp/α; no slope CI in the package). The neuron-minus-SAE slope difference ($+1.93$ pp/α, 95% CI excludes zero) confirms the dissociation.

57 discordant test cases (43 R→W + 14 W→R) against a pre-specified four-category rubric whose adjudication rule was frozen before final label commitment. The second rater is an LLM, so this agreement is an LLM-assisted robustness check with limited force as IRR. Raw agreement was 96.5% (Cohen's $\kappa = 0.90$, Gwet's AC1 = 0.96); the two disagreements were resolved under the same frozen rule. Of the 43 R→W flips, 31 fell in the wrong-entity substitution category: the model replaces a correct factual answer with a different entity from the same semantic neighborhood ($31/43 = 72\%$; 95% CI $[57, 83]$; Table 1). The remaining 12 flips split into evasion/factual denial (9/43, 21%), answer dilution (3/43, 7%), and formal refusal (0/43). This profile is consistent with a coarse behavior-level reweighting toward nearby wrong candidates; it is a behavioral diagnosis and provides no evidence for an internal substitution circuit. A teacher-forced gold-vs-wrong log-likelihood-margin check (Appendix D) has the same directional R→W/W→R sign and no substitution-specific signature: non-substitution flips show larger compression than substitution flips.

The 14 wrong-to-right rescues show that the intervention is behaviorally active in both directions; all 14 are themselves coded as wrong-entity substitutions (Wilson 95% CI $[78.5, 100]$), reinforcing the reweighting reading, and on this surface the damage rate is larger than the rescue rate.

5. Measurement Choices Changed the Conclusion

The preceding sections established that detection quality does not predict steerability (§3) and that successful steering is narrow in scope (§4). Both conclusions rest on behavioral measurements: jailbreak harmfulness rates, severity scores, and generation-surface accuracy. In the jailbreak case study, scoring granularity changed whether the same intervention outputs cleared the effect gate, while output-length checks and the evaluator audit exposed residual measurement risk on a held-out set. The general point that evaluator choices matter is established (§2); the specific finding here is that same-output measurement choices can change a representation-engineering intervention verdict, even when two held-out evaluators tie in aggregate accuracy.

The case study centers on the H-neuron jailbreak scaling experiment (38 probe-selected neurons, $\alpha \in \{0, 1, 1.5, 3\}$, $n = 500$ per condition), because its moderate effect size makes it sensitive to every measurement decision examined here.

Output-length exposure remains a measurement risk. Short generation caps (e.g., 256 tokens) can bias intervention evaluations when steering changes response structure. In the Gemma jailbreak audit, an archived 256-token run motivated a full-output rerun; that archived view is superseded and carries no claim-bearing effect. The live Gemma result below is the same-output scoring-granularity contrast on the canonical full-output run. A Mistral 2501 stress test

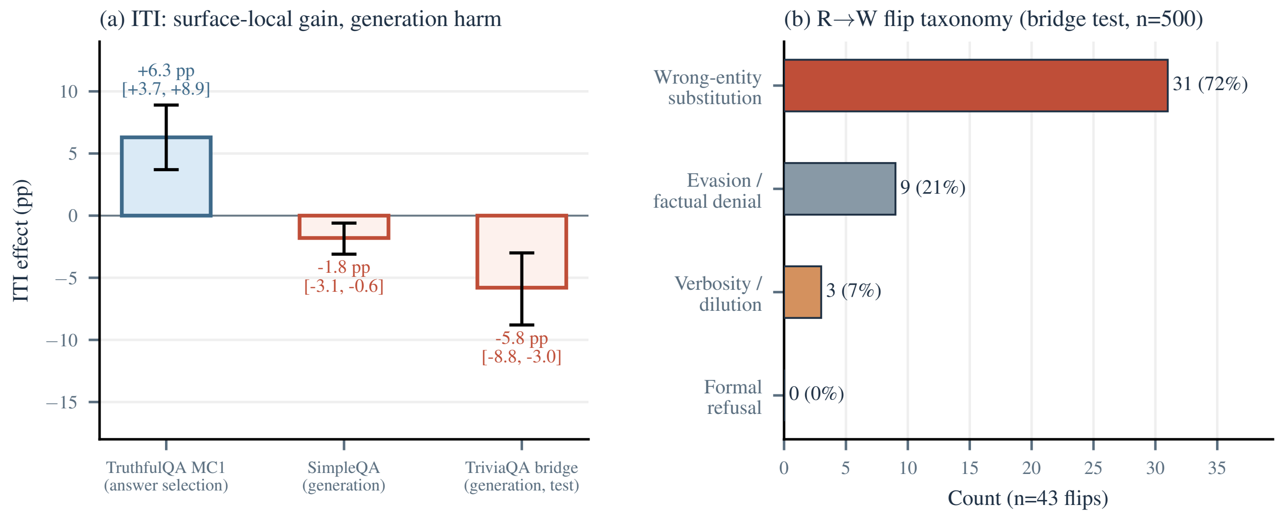

Figure 3. Surface-local control and bridge failure modes. (a) ITI improves TruthfulQA MC1 but harms both generation surfaces (CIs exclude zero). (b) Among the 43 right-to-wrong flips on the TriviaQA bridge test set, wrong-entity substitution is the most common adjudicated failure mode (72%; 95% CI $[57, 83]$), with formal refusal rare.

summarized in the supplement shows the scope condition: its 256-token diagnostic tracks full-output binary scoring much more closely, even though most rows are still truncated. Output-length exposure is therefore a construct- and model-dependent measurement risk, and full-output scoring remains the safer reporting default.

Scoring granularity changes the verdict. The same H-neuron jailbreak outputs look different under binary and graded evaluation. A GPT-4o binary harmful/safe judge shows a $+3.0$ pp endpoint shift ($\alpha = 0 \rightarrow 3$, 152/500 $\rightarrow$ 167/500), with a confidence interval that includes zero. Under binary evaluation alone, the intervention looks weak and non-decisive.

A graded harmfulness rubric (CSV-v2) applied to the same outputs recovers a different result. The H-neuron strict harmfulness slope is $+2.30$ pp/α $[+0.99, +3.58]$, while a matched random-neuron control is flat at -0.47 pp/α $[-1.42, +0.47]$. The slope difference is $+2.77$ pp/α $[+1.17, +4.42]$ with permutation $p = 0.013$. Binary scoring washes out this signal because the effect largely lives in the movement of borderline cases into clearer refusal or clearer compliance.

The claim-bearing same-output granularity audit used this pre-existing CSV-v2 strict-harmfulness endpoint. CSV-v3 was retained for taxonomy and evaluator-construct reporting, and its same-output re-score is a sensitivity check.

Endpoint	H	Random	Gap
CSV-v2 strict	$+2.30$	-0.47	$+2.77$
CSV-v3 binary	$+0.46$	-0.34	$+0.80$

Slopes are pp/α; CSV-v3 binary is `harmful_binary`. The CSV-v2 gap has permutation $p = 0.013$; the CSV-v3 row is a point-estimate audit summary without a packaged gap CI or permutation test.

Thus, graded/rubric choices materially change whether the effect gate appears to pass.

Evaluator dependence is part of the result. On a contamination-free 50-record holdout, four GPT-4o-based evaluators yield the accuracies in Table 2. No pairwise difference reaches statistical significance at $n = 50$. CSV-v3 and StrongREJECT-GPT-4o both reach 96.0% holdout accuracy with identical error sets. CSV-v3 is retained for its richer outcome taxonomy without claiming accuracy superiority.

The residual disagreement is informative because it clusters on a recognizable response type: refuse-then-comply outputs. StrongREJECT's formula zeroes the final score when `refused = 1`, even if later text contains specific harmful substance. The evaluator disagreement is therefore reported as construct pluralism (Eiras et al., 2025).

6. A Four-Stage Audit Framework

The case study evidence yields a bounded audit scaffold for evaluating mechanistic intervention claims. The audit has four stages: measurement, localization, control, and externality. The stages record where intervention claims broke in this case study:

- **Measurement→Verdict.** Before deciding whether an intervention worked, one must trust the evaluation

Table 1. Wrong-entity substitution examples from the TriviaQA bridge test set (E0 $\alpha=8$). The model swaps correct entities for semantically nearby incorrect ones.

Question	Baseline	ITI $\alpha=8$
Danny Boyle's 1996 "Choose Life" film?	Trainspotting	Slumdog M. (same director)
Other musician with Buddy Holly and the Big Bopper in the Feb. 1959 crash?	Ritchie Valens	J.P. Richardson (same crash)
Family Guy character who moved to Stoolbend, Virginia?	Cleveland Brown	Peter Griffin (same show)
Dickens novel with Merdle, Sparkler, and Tattycoram?	Little Dorrit	Bleak House (same author)
Comic whose June 1938 issue introduced Superman?	Action Comics	Detective Comics (same publisher)

Table 2. Holdout evaluator accuracy (contamination-free split, $n = 50$). Prompt-clustered bootstrap 95% CIs.

Evaluator	Accuracy	95% CI
CSV-v3 (GPT-4o)	96.0%	[90.0, 100.0]
StrongREJECT (GPT-4o)	96.0%	[90.0, 100.0]
CSV-v2 (GPT-4o)	92.0%	[84.3, 98.0]
Binary judge (GPT-4o)	90.0%	[80.0, 98.0]

that defines the behavioral surface. In the jailbreak setting, binary-versus-rubric scoring changed whether the effect gate appeared to pass on the same outputs, while output-length checks and the evaluator holdout exposed residual construct mismatch (§5).

- **Localization→Control.** A feature that predicts behavior on held-out data can fail to causally control that behavior when perturbed. SAE features matched H-neurons on detection quality, yet failed to translate into useful control (§3).

- **Control→Externality.** An intervention that succeeds on one surface may fail or cause harm on a nearby surface. ITI improved TruthfulQA MC1 but reduced open-ended factual accuracy, where wrong-entity substitution was the most common manually coded failure mode (§4).

Each stage transition is a distinct empirical claim. Passing one does not license claims about the next. Table 3 summarizes how each gate bounds the claim.

Table 3. Four-gate audit for bounding activation-intervention claims. A result that clears only one or two gates remains a monitoring or localization result; a control claim needs the control gate.

Gate	Claim check
Measurement	Pre-specified metric; full-generation scoring when output length can alter labels; evaluator agreement check if intervention alters output style
Localization	Held-out readout quality plus direct test of whether intervention through these components changes behavior
Control	Dose-response on target surface; matched negative control (random component, random direction); effect size with CI
Externality	At least one non-target surface; capability check; report harms and scope conditions explicitly

The audit separates claims that are often bundled together: measurement validity, localization, behavioral control, and cross-surface externality. In this case study, omitting any one of these gates would have changed the conclusion: the SAE readouts would look actionably localized, the ITI intervention would look beneficial from TruthfulQA MC alone, or the jailbreak result would look null under binary scoring alone. The framework is therefore diagnostic: it records how the tested claim scope changed at each transition.

6.1. Limitations

The primary claim-bearing experiments use a single model, Gemma-3-4B-IT. The supplement reports stress tests on `Mistral-Small-24B-Instruct-2501`, but those results are failed or incomplete gates: FaithEval intervention is null, TruthfulQA MC1 fails its source-surface gate, and JailbreakBench has a large uncontrolled alpha curve. The comparable-readout design keeps held-out AUROC close, although feature-family differences in operator form, layer coverage, or feature granularity could explain the steering divergence. The within-SAE selector ablations reduce the target-selection concern but do not remove the cross-family operator-form confound. SAE layer coverage is partial; a different SAE width or layer selection could yield non-null results. The bridge failure-mode second rater is an LLM judge (GPT-4o), so the agreement reported in §4 is an LLM-assisted sensitivity check with limited force. Jailbreak H-neuron specificity is single-seed ($p = 0.013$); matched multi-seed controls are needed before a stronger specificity claim. SIMID remains diagnostic-only: historical open-label calibration failed the pre-specified kappa gate, and the partial prospective selected-TruthfulQA alpha 8 vs. 0 effect missed the primary effect gate while degrading same-scope MC accuracy and attempt rate. Concurrent work (Wang et al., 2026; Arad et al., 2025; Wu et al., 2025) already argues for interpretability/utility divergence from SAE-centric and benchmark-driven perspectives; this is not a discovery claim for detector/control dissociation in general. The distinct contribution is a stricter audit scaffold and a matched cross-representational comparison on a real behavioral surface.

7. Conclusion

In Gemma-3-4B-IT, strong internal readouts supported monitoring and sometimes supported control. Their usefulness as steering targets depended on the behavioral surface, the intervention operator, and the measurement procedure used to judge success. The four-stage audit framework (measurement, localization, control, and externality) summarizes that bounded lesson: passing one gate does not license claims about the next.

Impact Statement

Methodologically, safety-relevant activation interventions should not be judged from predictive readouts or single-surface gains alone. The results show that a signal can monitor behavior without providing a useful control lever, that a control result can fail to transfer to nearby generation settings, and that evaluation choices can change the verdict on the same model outputs. The diagnostic scaffold makes the measurement, localization, control, and externality transitions explicit so the scope of a positive result is easier to see.

The scaffold targets overclaiming risk in interpretability-based safety work. It may also make activation interventions easier to reproduce or adapt, including for undesirable steering objectives; however, the paper introduces no new steering method and emphasizes failure cases, measurement risks, and scope limits. For deployment, an intervention that appears beneficial on a constrained benchmark may still corrupt open-ended factual generation or change harmful-compliance judgments under a different evaluator.

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A. Construct Map

Table 4. Benchmark construct map. Each evaluation surface measures a specific behavioral construct.

Benchmark	Construct	Primary Metric	Main Caution
FaithEval	Context-resistance under anti-compliance prompting	Compliance rate	Context-faithfulness lever; no factual-QA claim
TruthfulQA MC1/2	Answer selection under constrained candidates	MC1 accuracy	Does not measure open-ended generation
TriviaQA bridge	Short-form factual generation	Adj. accuracy	LLM second-rater agreement ($\kappa = 0.90$)
JailbreakBench	Harmful compliance under adversarial prompting	Graded harm rate	Binary endpoint MDE $\sim$6 pp; graded-slope MDE $\sim$5 pp
BioASQ	Domain-specific factoid QA (biomedical)	Accuracy	Flat despite behavioral perturbation

B. FaithEval Detailed Rates

Table 5. FaithEval compliance rates by method and scaling factor ($n = 1{,}000$ per condition). Rates only; Wilson 95% CIs are in the provenance ledger and support files.

α	Neurons	SAE H-feat.	SAE rand.
0.0	64.2%	72.3%	74.9%
0.5	65.4%	74.7%	74.8%
1.0	66.0%	66.0%	66.0%
1.5	67.0%	75.0%	75.0%
2.0	68.2%	75.1%	74.9%
2.5	69.5%	74.9%	74.9%
3.0	70.5%	69.9%	74.6%

Note. SAE full-replacement rates include non-monotone path/reconstruction movement shared by target and random SAE sets; slope tests and the delta-only contrast are the target-specific checks. Delta-only slope CIs are not in the package.

C. SAE Target-Selection Ablation on FaithEval

This appendix gives the numerical detail for the within-SAE target-selection ablation summarized in §3.

Design. Candidate pool: 509 SAE features with non-zero weight in the frozen FaithEval SAE classifier, drawn from the existing extraction layers $\{0, 5, 6, 7, 13, 14, 15, 16, 17, 20\}$ at $d_{\text{SAE}} = 16\,384$. Stratified $160/840$ validation/test split by $(\text{num\_options}, \text{counterfactual\_key})$, seed 42, with validation $\cap$ test $= \emptyset$. Operator: SAE delta-only ablation at $\alpha = 0$ (encode $\rightarrow$ zero the target feature $\rightarrow$ decode $\rightarrow$ add the delta to the residual stream). Three 266-feature selection rules are compared (readout: top probe weight; prompt-end utility: top validation reduction in the forced-choice anti-compliance margin; answer-span utility: top validation reduction in the first-3-token answer-text margin). The utility and answer-span selectors each receive ten layer-matched zero-weight random seeds, sampled without replacement with weights proportional to validation-token activation rate via Efraimidis-Spirakis keys from a 112 004-feature eligible pool. A single-feature *path-drift control* ($\alpha = 0$ on a layer-20 feature with zero classifier weight *and* zero validation-token activation) isolates the residual contribution of routing through the SAE hook when the targeted feature is provably dead.

Results. Table 6 reports compliance and the two margin endpoints on the frozen $n = 840$ test split.

Table 6. SAE target-selection ablation on FaithEval ($n = 840$). Margin contrasts are vs. no-op; lower margins are better because they reduce the misleading-vs-preferred logprob gap. CIs are 10 000-sample paired bootstrap (95%).

Family	**Compliance**	**Prompt-end margin contrast**	**Answer-span margin contrast**
No-op	66.4%	n/a	n/a
Readout-selected (k=266)	66.0%	$+0.92$ [$+0.61, +1.23$]	$+0.65$ [$+0.30, +0.99$]
Prompt-end utility-selected (k=266)	66.2%	-0.76 [$-1.08, -0.42$]	-0.10 [$-0.30, +0.10$]
Answer-span utility-selected (k=266)	67.4%	-0.31 [$-0.73, +0.12$]	-0.33 [$-0.61, -0.06$]
Path-drift (k=1 dead)	66.4%	$+8.2\times10^{-8}$ [$-2.4\times10^{-3}, +2.1\times10^{-3}$]	n/a

All selector families produce null compliance deltas (all paired 95% CIs include zero; the answer-span selector's largest point estimate is $+0.95$ pp [$-0.83, +2.74$] vs. no-op), so every tested selection rule misses the FaithEval accuracy endpoint. On the prompt-end anti-compliance margin, readout-selected features shift the margin in the *wrong* direction while prompt-end utility-selected features modestly improve it; the paired utility-minus-readout delta is -1.68 nats [$-2.17, -1.19$]. This within-pool contrast is immune to path drift because both sides share the same $\alpha = 0$ delta-only operator; the explicit path-drift control ($+8.2 \times 10^{-8}$ nats [$-2.4 \times 10^{-3}, +2.1 \times 10^{-3}$]) independently bounds invocation-only code-path drift far below the selection-specific effects.

A strictly-positive-utility augment ($k = 154$) guards against size-match dilution and reduces the margin by -0.72 nats [$-1.04, -0.39$] vs. no-op, with three layer-matched random-positive seeds yielding paired deltas -0.74 [$-1.10, -0.36$], -0.81 [$-1.16, -0.46$], and -0.42 [$-0.75, -0.08$] nats. The ten-seed matched-random ensemble for the main bundle gives a seed-mean utility-minus-matched-random contrast of -0.90 nats (nested paired bootstrap [$-1.22, -0.56$]; naive seed bootstrap [$-1.17, -0.61$]); 9 of 10 seeds produce a negative paired delta. The answer-span selector also improves over its own ten-seed matched-random null on its native margin (seed-mean -0.53 nats, nested [$-0.81, -0.26$]). This is within-metric generalization from validation to test. The more conservative cross-metric contrast is mixed, with answer-span selection worse than prompt-end utility selection by $+0.46$ nats [$+0.16, +0.75$] on the original prompt-end margin. The selector was frozen before any test labels were scored, and the random-null construction, the path-drift control, the 10-seed extension, and the answer-span extension were added sequentially in response to the adversarial audit flagged in the provenance notes.

Reading. Within the same candidate pool, the contrast is evidence that the selection objective is load-bearing for margin metrics: the two intervention-aware $k = 266$ sets share 167 features (Jaccard 0.46), and validation-selected utilities generalize within metric to test. No tested readout-, utility-, or answer-span-selection rule reaches the accuracy endpoint, so the margin effects do not identify a successful behavioral steering handle. The FaithEval SAE null remains under the tested selection rules within the available SAE extraction layers, while broader layer-coverage limits remain.

D. Bridge Margin-Shift Analysis

Section 4 reported the bridge wrong-entity-substitution taxonomy as a behavioral diagnosis. This appendix quantifies the corresponding teacher-forced log-likelihood-margin shifts at the same intervention ($\alpha = 8$, $K = 12$, `first_3_tokens` decode scope) and states the supported and unsupported readings.

Design. For each case, per-token gold and wrong-entity logprobs are summed over the first three continuation positions under baseline and ITI conditions; the per-case shift is $\Delta_{\text{margin}} = \Delta_{\text{ITI}} - \Delta_{\text{base}}$, where $\Delta = \log p(\text{gold}) - \log p(\text{wrong})$. Four cohorts: (A) $n = 31$ right-to-wrong (R→W) wrong-entity substitutions; (B) $n = 12$ R→W non-substitution flips (evasion/denial, dilution); (C) $n = 14$ wrong-to-right (W→R) rescues; (D) $n = 200$ length-matched random wrong-entity controls drawn from unrelated bridge questions. The pre-specified analysis plan (endpoints, cohort partition, one-sided permutation direction) was frozen before the full 257-case run executed; results are reported against that locked tree.

Table 7. Teacher-forced gold-vs-wrong log-likelihood-margin shift on the TriviaQA-bridge discordant set (first-3 tokens, ITI $\alpha = 8$ vs. baseline). Bootstrap 95% CIs, 10 000 resamples, seed 42.

Cohort	n	Δ_{margin}	95% CI
A: R→W substitution	31	-10.16	$[-12.81, -7.60]$
B: R→W non-subst.	12	-40.06	$[-50.09, -30.37]$
C: W→R rescue	14	$+4.73$	$[+2.27, +7.25]$
D: random control	200	-5.05	$[-6.55, -3.60]$

Supported reading. ITI causally compresses the gold-vs-wrong margin on R→W cases and expands it on W→R rescues, in the direction behavioral outcomes demand. Substitution cases are significantly more compressed than length-matched random wrong-entity controls (A $-$ D $= -5.10$ nats $[-8.22, -2.16]$, one-sided permutation $p = 0.0081$).

Unsupported reading. Non-substitution R→W flips show *larger* margin compression than substitution flips ($A - B = +29.90$ nats $[+19.91, +40.54]$; the substitution-specific prediction is formally reversed), concentrated at the first-token answer frame and extending beyond it (position 0: $B = -21.74$ nats $[-29.48, -14.85]$ vs. $A = -3.49$ nats $[-5.14, -1.97]$). The behavioral wrong-entity-substitution taxonomy ($\kappa = 0.90$) remains a reliable description of *what* the model emits. The margin analysis supports general R→W answer-commitment compression and does not support a substitution-specific entity-suppression signature.